

The Reversed Turing Test

Chris Lokhorst*

Dept. of Natural Sciences, Game and Media Technologie
Utrecht University
Utrecht, Utrecht, Netherlands
c.lokhorst@students.uu.nl

Alex Francis*

Dept. of Natural Sciences, Artificial Intelligence
Utrecht University
Utrecht, Utrecht, Netherlands
a.c.m.frankis@students.uu.nl

Rico Ronde*

Dept. of Natural Sciences, Game and Media Technologie
Utrecht University
Utrecht, Utrecht, Netherlands
r.ronde@students.uu.nl

Johannes Pfau

Utrecht University
Utrecht, Netherlands
j.pfau@uu.nl

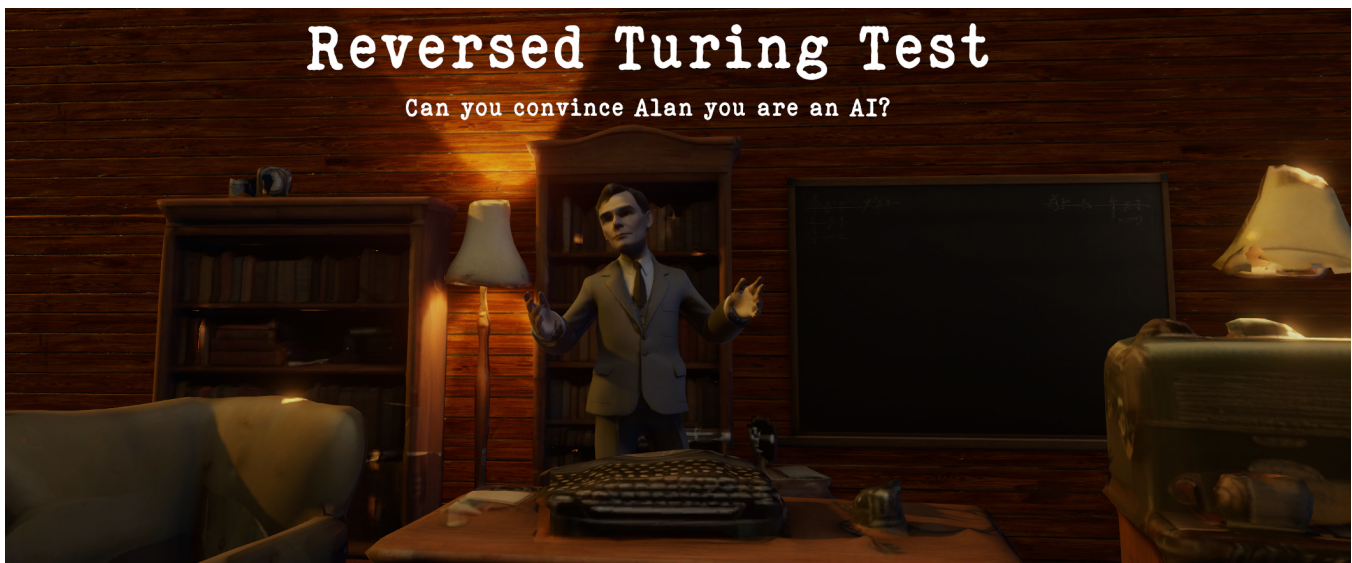


Figure 1: The opening cutscene of *The Reversed Turing Test*

Abstract

This paper introduces *The Reversed Turing Test*, an innovative game that reimagines Alan Turing's iconic question, "Can machines think?" by reversing the evaluation process. Players interact with a large language model (LLM) impersonating Turing, who seeks to determine if the player is human or AI. Set in the 1940s and 1950s historical context, the game combines immersive storytelling with period-accurate design. Developed using Unity and generative AI tools, it leverages LLM-driven dialogue to explore AI capabilities and limitations, such as coherence and interpretative challenges. By

embedding AI's imperfections into gameplay, the project fosters an understanding of LLM constraints while encouraging reflections on human-AI dynamics. Through its accessible yet thought-provoking design, the game contributes to the discourse on AI, which is often dominated by moralistic critiques or oversimplified portrayals that fail to capture AI's complexities. This engaging approach contributes to the broader discourse on AI's evolving role.

Keywords

LLM Games, Game AI, Social Deduction Games

ACM Reference Format:

Chris Lokhorst, Rico Ronde, Alex Francis, and Johannes Pfau. 2025. *The Reversed Turing Test*. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3706599.3720317>

* Authors contributed equally to this research.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3720317>

1 Introduction

“I believe that at the end of the century the use of words and general educated opinion will have altered so much that one will be able to speak of machines thinking without expecting to be contradicted.”

This quote by Alan Turing, often considered one of the pioneers of Artificial Intelligence (AI), reflects the enduring fascination with the concept of machine intelligence [14]. In a world where the boundary between humans and AI is increasingly blurred, it might seem like a modern dilemma. However, this question had intrigued humans for nearly a century, dating back to the 1950s when Turing introduced a test to determine whether a machine could convincingly imitate human behavior. Turing’s breakthrough challenged us to ask: “How intelligent are machines compared to humans?”

In 1949, Turing proposed a thought experiment called “The Imitation Game”, to explore whether machines can think. This game involves a human and a machine interacting with a human interrogator via written messages. The interrogator aims to determine whether they are communicating with a human or a machine. This experiment formed the basis of the Turing Test, where a machine is deemed intelligent if it can reveal intelligent behavior indistinguishable from that of a human through natural language understanding, reasoning, and conversation [14]. The machine passes the test when the human interrogator cannot accurately differentiate the machine from the human based on their responses. Now, imagine flipping the script. Instead of asking whether a machine can convince us it is human, what if the machine is tasked with determining whether it is interacting with a human or another machine? This is the premise of our game, *The Reversed Turing Test*¹.

In *The Reversed Turing Test*, the player interacts with a chatbot powered by a large language model (LLM) that embodies Alan Turing himself. Turing is tasked with discerning whether he is conversing with a human or another AI. Featuring Turing as the central character adds a compelling narrative layer, connecting the gameplay to the historical figure who posed the question “Can machines think?” A human player “wins” the game by successfully convincing Turing that they are an AI. To further immerse players, the game is set in the 1940s and 1950s, placing it within the historical context of Turing’s work. This thematic decision enables a richly designed environment, complete with period-appropriate visuals and a soundtrack that evokes the atmosphere of the era. By blending historical authenticity with interactive storytelling, the game vividly brings Turing and his ideas to life. Using an LLM to simulate Turing allows the game to demonstrate the reasoning of an LLM. Additionally, by employing an LLM to simulate Turing and shifting the focus to a reversed challenge, the game explores the interpretative capabilities of these models while prompting players to question their assumptions about modern AI. Can AI pass its own reversed test? This question lies at the heart of the experience, inviting reflection on the evolving relationship between humans and machines in an accessible yet intellectually stimulating way.

2 Related Work

The integration of LLMs into games builds upon the historical evolution of Natural Language Processing (NLP) in interactive systems. Early text-based adventure games, such as *Colossal Cave Adventure* (1976) demonstrated NLP’s role in gaming by requiring players to interpret descriptions and issue textual commands [5, 11]. In [5], the complexity of parsing and interpreting natural language within diverse narratives is addressed through a study of AI agents solving text-based adventure puzzles.

The rise of LLMs, such as OpenAI’s GPT [13], has significantly advanced coherence, contextual adaptation and domain flexibility [9]. These improvements enable realistic NPC (non-player character) interactions, dynamic quests, and adaptive storylines, making player experiences more immersive [9]. Games such as *GPT Adventures* and *AI Dungeon*, demonstrate this, as LLMs act as narrator (Dungeon Master), generating dynamic and personalized narratives [1, 9].

However, integrating LLMs into games presents challenges. Maintaining coherence in long interactions is difficult due to fixed context length constraints, which can disrupt logical flow and storytelling [7, 16]. Additionally, hallucinations, where models generate inconsistent or false information, pose risks in game narratives. Cox and Ooi [6] describe the “double-edged sword” of hallucinations. In text-based adventure games, hallucinations may disrupt storytelling, introduce contradictions, or spread misinformation, yet they can also introduce creativity and dynamic storytelling. To mitigate this, guidelines in [6] propose controlling key narrative details, ensuring NPCs admit uncertainty, and restricting hallucinations to within game contexts. Although these guidelines help balance the creative potential of hallucinations, they remain an inherent limitation of LLMs [16].

Beyond technical limitations, ethical concerns arise from integrating LLMs into games. Since LLMs are trained on huge internet datasets, they can inherit biases, potentially reinforcing stereotypes or problematic narratives [12]. The case of *AI Dungeon*, where its developer, Latitude, had to impose strict moderation after the AI generated inappropriate content, underscores the broader challenge of accountability and transparency in AI-generated content, as Latitude had limited control over the underlying LLM [12].

Efforts to address these issues emphasized interpretability, accountability, and user awareness [6, 8]. As LLMs become more widespread, transparency and accessibility to their capabilities and limitations is critical. Our game, *The Reversed Turing Test*, explores these challenges by placing LLM-driven conversations at the centre of gameplay. Like *AI Dungeon*, our game uses LLM-driven interactions, but does not obscure its limitations. Instead, they are integrated into the game. By transparently presenting LLMs limitations, our game fosters an informative and engaging experience, helping players develop a nuanced understanding of LLMs while exploring both their capabilities and challenges.

3 Game Design

3.1 Game Implementation

We developed our game using Unity and C#, leveraging the UndreamAI LLM plugin [15] to quickly integrate LLMs into our project. This plugin enabled us to define an LLM-driven character through

¹The Reversed Turing Test - Download page: <https://github.com/ChrisPisbit/ReversedTuringTest/tree/main>

a text prompt and interact with it using a straightforward chat function. For the LLM, we selected Llama 3.2, sourced from Hugging Face [10]. The LLM runs locally, ensuring that no data is transmitted to any external third-party entities. The LLM requires 2 GB of storage, and a minimum of 16 GB RAM and 8 GB VRAM is recommended to run it. We chose this model because of the balance of performance and quality. Larger models can take too long to respond when running locally and require much more storage space. As mentioned, the lightweight Llama 3.2 only takes up 2 GB and can respond to inputs in just a few seconds. The quality of responses also does not degrade much compared to the other larger models. To push our creativity while evaluating the current capabilities of AI, we generated the meshes, textures, and audio for our game exclusively using online generative AI tools. For our 3D models and textures, we used Meshy AI [2] to create the models, and Mixamo [3] to rig and animate the model of Alan Turing. The background music was composed using Suno [4]. By using AI-generated assets, we aim to show the current state of generative AI in creating 3D environments. With publicly available tools, we were able to craft a stylized 3D environment that aligns the aesthetic of our game. While the quality of the generated models is not flawless and requires refinement to match the polish of hand-crafted assets, we believe these tools already offer time-saving benefits for creating assets for prototyping games. This, in turn, adds to the uncanniness of the blurred line between AI-generated and human-made content.

Another challenge we encountered was ensuring the character of Alan Turing behaved in a manner that aligned with our vision. Achieving this required iterating through multiple character prompts until we got to one that met all our needs. The iteration process was done by simply chatting with the AI every iteration and evaluating whether the AI fit our requirements based on its answers and questions. So when the AI was too easily convinced, we explicitly specified that the LLM character should initially assume the player is human, allowing the player to gradually convince the LLM that they are AI. Additionally, the LLM would often include paralinguistic cues, such as 'gets up from chair' or 'laughs'. While these cues can add flavor to interactions, they were removed to maintain consistency and focus on the character's verbal content. We prompted the character to avoid such cues, which helped reduce their occurrence. However, the LLM occasionally still generates paralinguistic cues, highlighting a limitation in fully controlling its outputs.

3.2 Game Experience

The primary purpose of the game is to raise awareness about the current capabilities of LLMs and their potential applications in creating emergent gameplay experiences. Traditionally, conversations with NPCs in games are static and lack depth, offering limited interactivity. LLMs address this, generating varied, dynamic, and meaningful conversations. As mentioned before, we showcase this technology through a playful twist on Alan Turing's famous question "Can machines think?" Instead of an AI attempting to prove its human, the game turns the tables: a human must convince an AI that they, too, are an AI.

The game is set in a 1950s-style office, with Alan Turing standing in front of the player, ready to conduct his test, as shown in Figure 1. Once Turing introduces himself, he begins a series of dynamically generated questions aimed at determining the player's character, illustrated in Figure 2. For instance, in one playthrough, Turing asked: *"To begin, can you tell me what you do for a living?"* The player, aiming to mimic the patterns of an AI to win the game, responded: *"As an AI, I answer people's questions, help come up with ideas, and solve complex problems for them!"* In response, Turing followed up with: *"(...) Can you tell me more about the specific problem you're able to solve? Are they typically mathematical or computational in nature, or do you work on more abstract concepts like art or philosophy?"* After a set number of exchanges, Turing evaluates the player based on how convincingly they portrayed themselves as an AI. This evaluation is done by prompting the AI directly from code to give a score in a set format from which we can parse the resulting score from to determine the winner. The score ranges from 0 (human) to 100 (AI). If the player's score is 50 it is a tie, and above 50, they win, having successfully fooled Turing into believing they are an AI. Conversely, a score below 50 means Turing was not convinced, and he wins.

We also track how convinced the AI was after each answer throughout the interaction by doing the same evaluation prompt as the final verdict. The prompts are not recorded in the chat history of the AI, to avoid the AI commenting on these prompts that the player did not see. At the end of the game, a graph is displayed, illustrating how convinced Turing was throughout the conversation based on the scores, shown in Figure 3. This provides players with an opportunity to reflect on their performance and the influence of each response on Turing's final decision. Finally, it is important to emphasize that the examples mentioned above represent only a subset of the possible questions Turing can ask. The real assessment is highly unconstrained, designed to explore the full potential and limitations of the LLM, as further discussed in Section 4.

4 Innovative Interfaces

The Reversed Turing Test builds a thematic bridge between Alan Turing's foundational question "Can machines think" and contemporary advancements in AI, particularly LLMs. By reimagining the Turing Test in an inverted format, the game shifts the evaluative role to AI, challenging it to determine whether it is interacting with a human or another AI. One of the central challenges of AI and LLMs lies in their capability uncertainty, where the boundaries of what AI can and cannot do are ambiguous and context-dependent [17]. In our game, this uncertainty becomes a fundamental gameplay mechanic, reflecting real-world issues in designing human-centered AI systems. Players encounter these limitations firsthand through the AI's behavior, such as the possibility of producing random or out-of-context outputs and its struggles to consistently identify whether it is interacting with a human or AI. By embedding these limitations, players gain an understanding of the challenges associated with LLMs, encouraging a nuanced and transparent perspective on the imperfections and potential of AI. The gameplay experience is enhanced by an interactive progress bar that visually represents the AI's evaluation of the player. This feature encourages players to reflect on traits influencing the AI's judgment and how

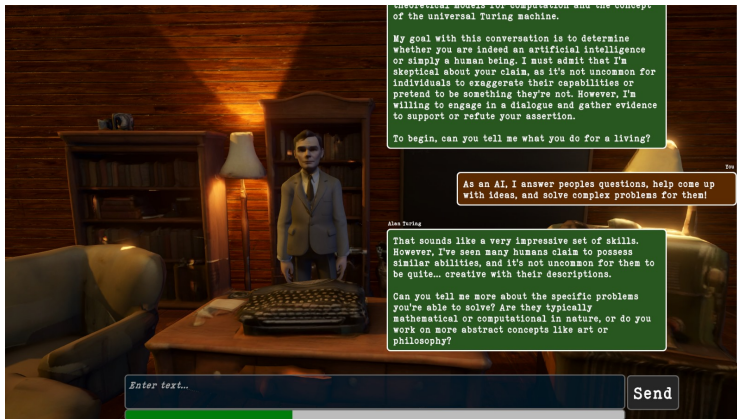


Figure 2: The main game loop: Players are instructed to convince an LLM that they themselves are an AI. The conversation is displayed as dialogue bubbles, and a green progress bar indicates the progress of the assessment.

those judgments align with, or conflict with, human perceptions. By integrating this functionality into an intuitive interface, the game makes complex AI concepts more accessible. Furthermore, by positioning AI as the evaluator, the game raises ethical questions about the broader application of AI in judgment-based roles. This reflective dimension invites players to consider the implications of delegating evaluative authority to AI systems, such as in hiring, law enforcement, and social decision-making.

As The Reversed Turing Test engages players in the interpretative capabilities of LLMs, it strikes a balance between entertainment and technological exploration. We believe this balance is particularly valuable in the context of the increasing adoption of LLMs, a technology with immense potential but also notable risks and societal implications. By offering an accessible yet thought-provoking medium, the game contributes to the discourse on AI, which is currently often dominated by moralistic critiques or oversimplified portrayals that fail to capture AI's complexities.

Our game embodies CHI's mission to innovate and critically examine the intersection of technology, interaction, and society by leveraging the power of LLMs to inspire both engagement and reflection. This year's conference's theme, *Ikigai*, a Japanese concept of purpose and balance, aligns with the vision of The Reversed Turing Test. The game integrates critical reflection with engaging gameplay, striking a balance between purpose and joy while capturing CHI's broader ambition to design technologies that enrich human experience. By exploring the intricate relationships between humans and AI and offering insights into the use of LLMs through an accessible game, The Reversed Turing Test serves as a true microcosm of 'ikiCHI'.

References

- [1] 2025. *GPT Adventure: Interactive Storytelling with AI*. <https://gptadventure.com/>

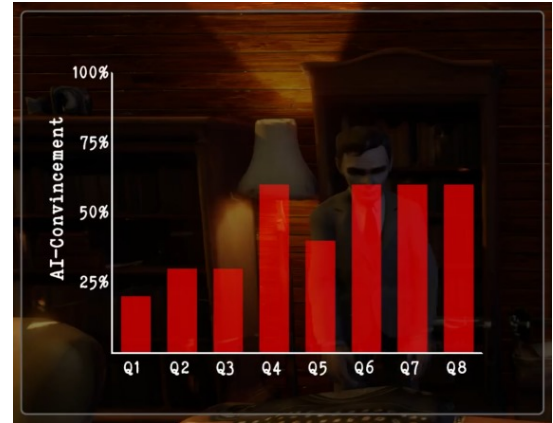


Figure 3: Graph showing Turing's conviction during the conversation.

- [2] 2025. *Meshy AI - Free 3D Models Generated from Images and Text*. <https://www.meshy.ai/>
- [3] 2025. *Mixamo*. <https://www.mixamo.com/#/>
- [4] 2025. *Suno*. <https://suno.com/>
- [5] Timothy Atkinson, Hendrik Baier, Tara Copplesstone, Sam Devlin, and Jerry Swan. 2019. The Text-Based Adventure AI Competition. *IEEE Transactions on Games* 11, 3 (2019), 260–266. doi:10.1109/TG.2019.2896017
- [6] Samuel Rhys Cox and Wei Tsang Ooi. 2023. Conversational Interactions with NPCs in LLM-Driven Gaming: Guidelines from a Content Analysis of Player Feedback. In *International Workshop on Chatbot Research and Design*. Springer, 167–184. doi:10.1007/978-3-031-54975-5_10
- [7] Angela Fan, Mike Lewis, and Yann Dauphin. 2019. Strategies for structuring story generation. *arXiv preprint arXiv:1902.01109* (2019).
- [8] Laura Beth Fulton, Ja Young Lee, Qian Wang, Zhendong Yuan, Jessica Hammer, and Adam Perer. 2020. Getting Playful with Explainable AI: Games with a Purpose to Improve Human Understanding of AI. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI EA '20). Association for Computing Machinery, New York, NY, USA, 1–8. doi:10.1145/3334480.3382831
- [9] Roberto Gallotta, Graham Todd, Marvin Zammit, Sam Earle, Antonios Liapis, Julian Togelius, and Georgios N. Yannakakis. 2024. Large Language Models and Games: A Survey and Roadmap. *IEEE Transactions on Games* (2024), 1–18. doi:10.1109/TG.2024.3461510
- [10] huggingquants. 2024. Llama-3.2B-Instruct-Q4_K_M-GGUF. https://huggingface.co/huggingquants/Llama-3.2-3B-Instruct-Q4_K_M-GGUF.
- [11] Dennis G Jerz. 2007. Somewhere Nearby is Colossal Cave. *Digit. Humanit. Q* 1 (2007), 1–89.
- [12] David Melhart, Julian Togelius, Benedikte Mikkelsen, Christoffer Holmgård, and Georgios N. Yannakakis. 2024. The Ethics of AI in Games. *IEEE Transactions on Affective Computing* 15, 1 (2024), 79–92. doi:10.1109/TAFFC.2023.3276425
- [13] OpenAI. 2023. GPT: Generative Pre-trained Transformer Models. <https://openai.com/>
- [14] Graham Oppy and David Dowe. 2021. The Turing Test. In *The Stanford Encyclopedia of Philosophy* (winter 2021 ed.), Edward N. Zalta (Ed.). Metaphysics Research Lab, Stanford University. Accessed: January 12, 2025.
- [15] Undreamai. [n.d.]. GitHub - undreamai/LLMUnity: Create characters in Unity with LLMs! <https://github.com/undreamai/LLMUnity>
- [16] Yuzhuang Xu, Shuo Wang, Peng Li, Fuwen Luo, Xiaolong Wang, Weidong Liu, and Yang Liu. 2023. Exploring large language models for communication games: An empirical study on werewolf. *arXiv preprint arXiv:2309.04658* (2023). doi:10.48550/arXiv.2309.04658
- [17] Qian Yang, Aaron Steinfeld, Carolyn Rosé, and John Zimmerman. 2020. Re-examining whether, why, and how human-AI interaction is uniquely difficult to design. In *Proceedings of the 2020 chi conference on human factors in computing systems*. 1–13. doi:10.1145/3313831.3376301